Memory Hierarchy (I)

Outline

- Random-Access Memory (RAM)
- Nonvolatile Memory
- Data transfer between memory and CPU
- Hard Disk
- Data transfer between memory and disk
- SSD
- Storage trends
- Suggested Reading: 6.1

Random-Access Memory (RAM)

- Key features
  - RAM is packaged as a chip.
  - Basic storage unit is a cell (one bit per cell).
  - Multiple RAM chips form a memory.

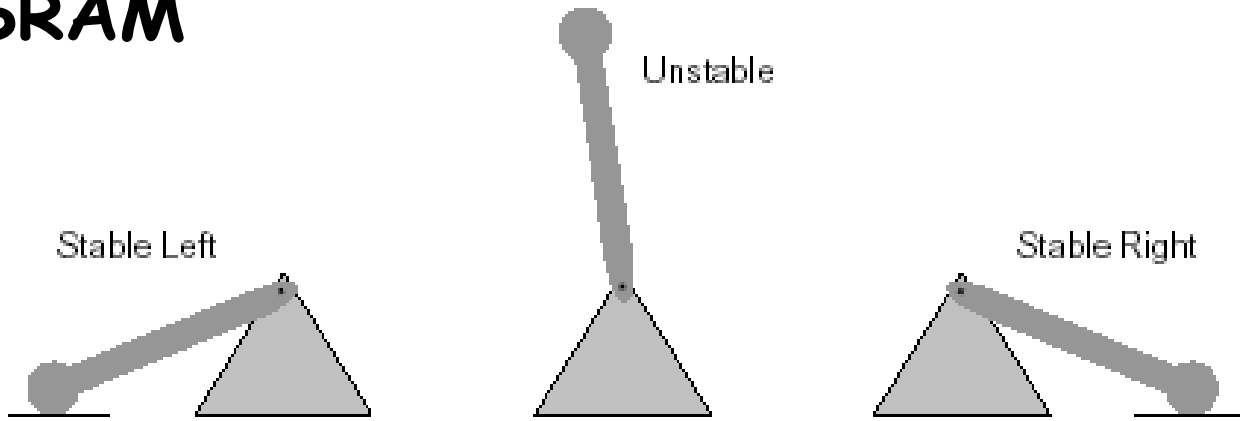
Random-Access Memory (RAM)

- Static RAM (SRAM)
  - Each cell stores bit with a six-transistor circuit.
  - Retains value indefinitely, as long as it is kept powered.
  - Relatively insensitive to disturbances such as electrical noise.
- Dynamic RAM (DRAM)
  - Each cell stores bit with a capacitor and transistor.
  - Value must be refreshed every 10-100 ms.
  - Sensitive to disturbances.

SRAM vs DRAM summary

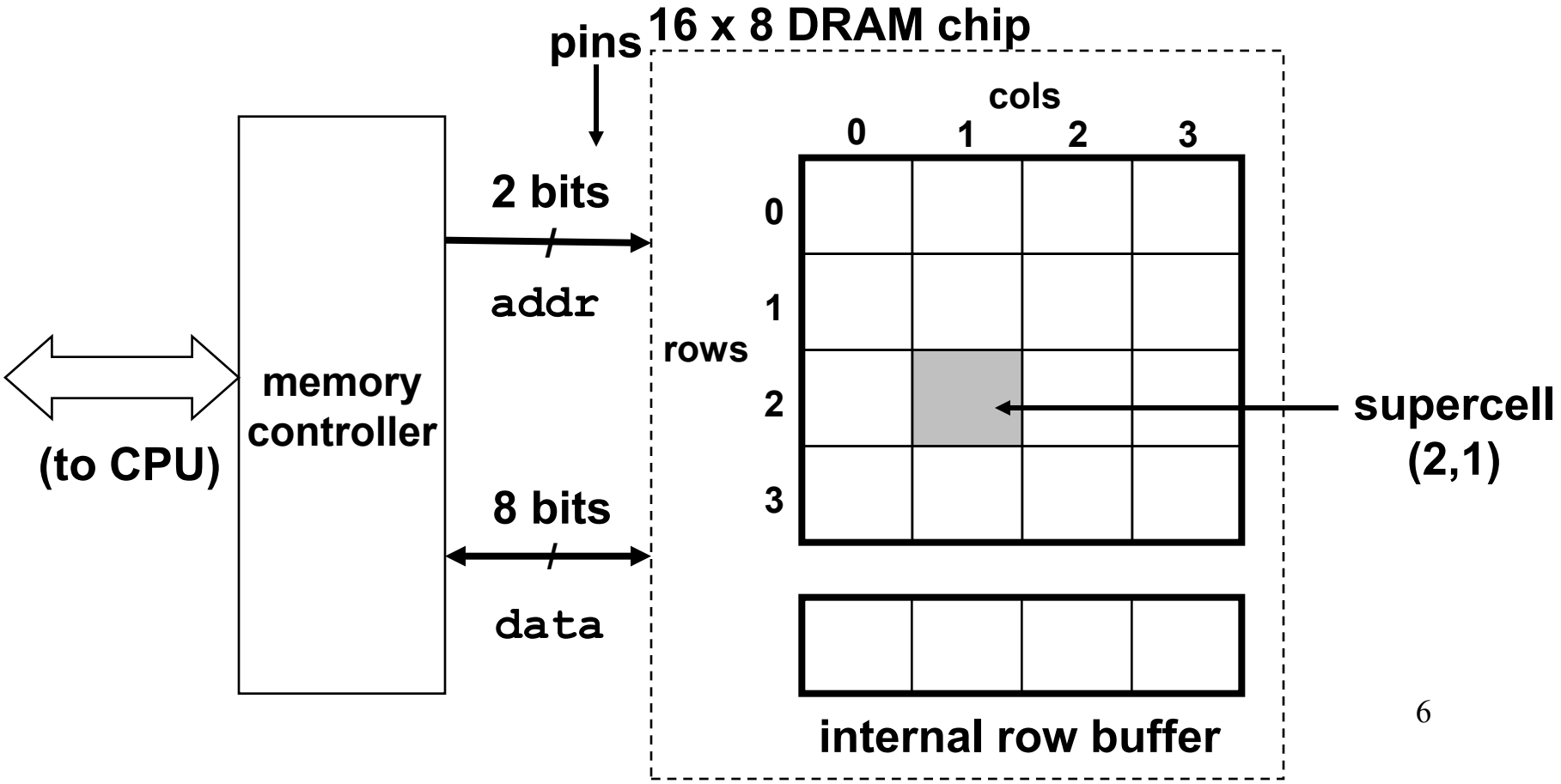
	Tran. per bit	Access time	Persist?	Sensitive?	Cost	Applications
SRAM	6	1X	Yes	No	1000x	cache memories
DRAM	1	10X	No	Yes	1X	Main memories, frame buffers

SRAM



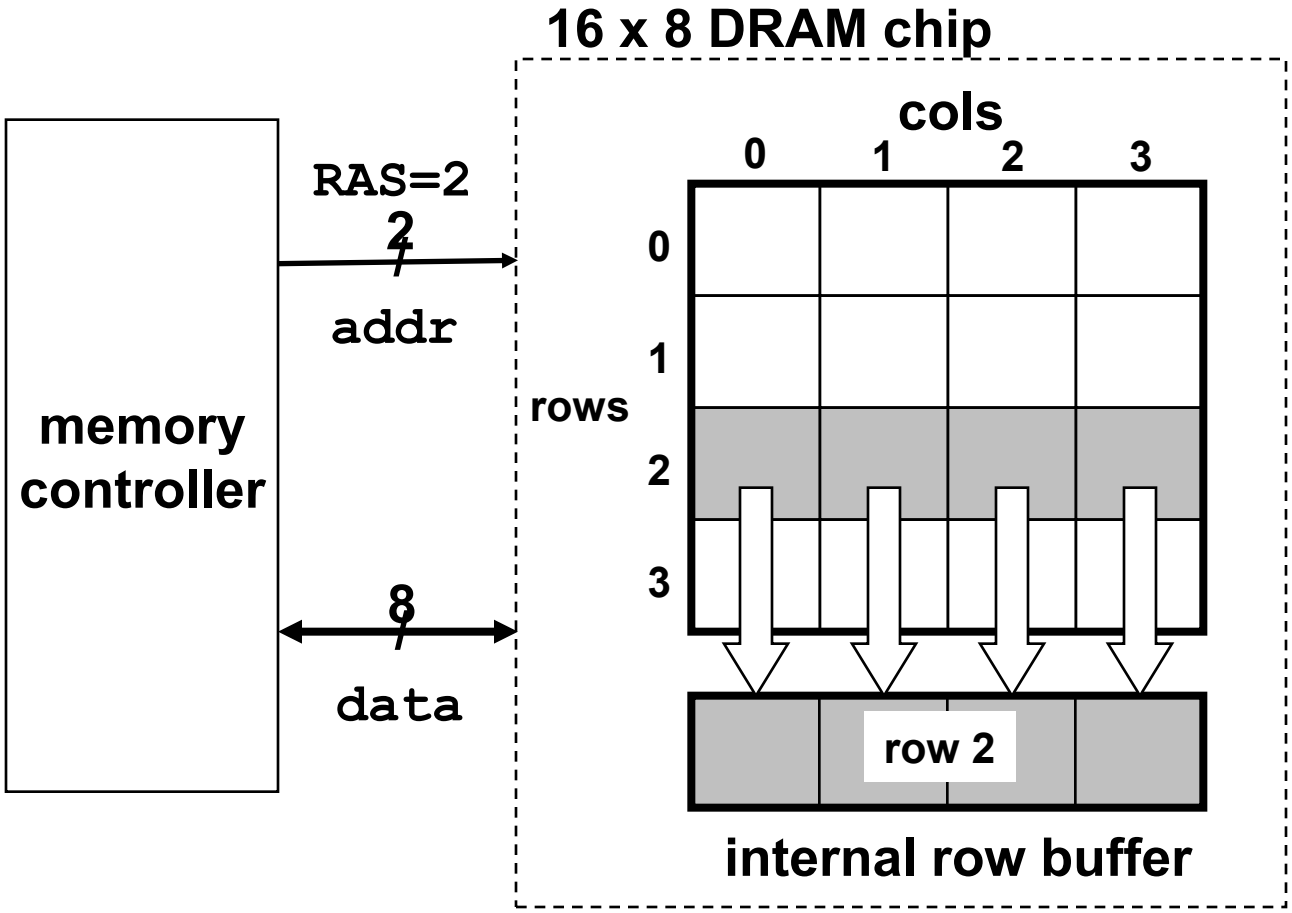
Conventional DRAM organization

- *d x w* DRAM:
  - dw total bits organized as *d* supercells of size *w* bits



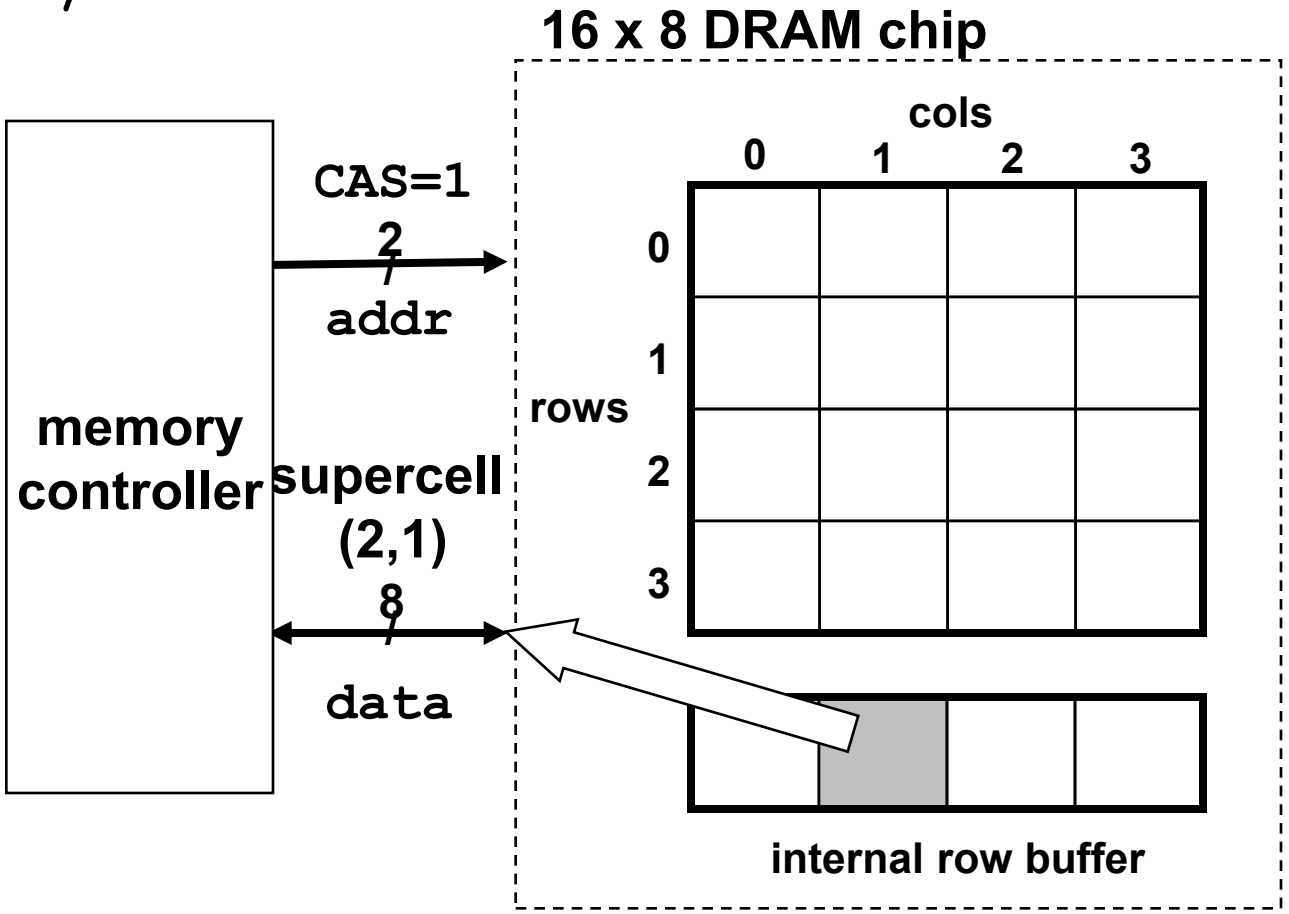
Reading DRAM supercell (2,1)

- Step 1(a): Row access strobe (RAS) selects row 2.
- Step 1(b): Row 2 copied from DRAM array to row buffer.

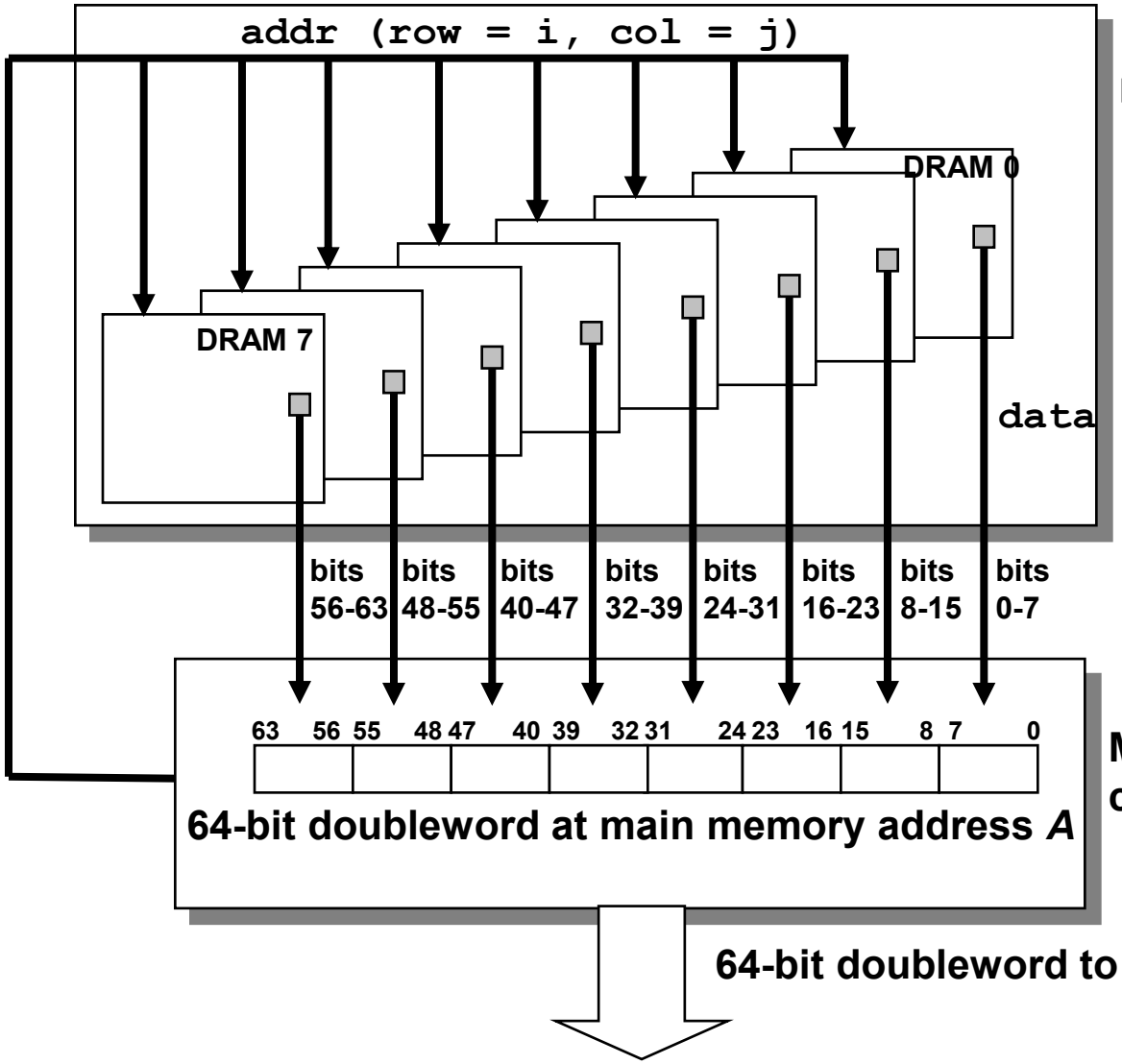


Reading DRAM supercell (2,1)

- Step 2(a): Column access strobe (CAS) selects column 1.
- Step 2(b): Supercell (2,1) copied from buffer to data lines, and eventually back to the CPU.



Memory modules



□ : supercell (i,j)

64 MB
memory module
consisting of
eight 8Mx8 DRAMs

DIMM: dual in-line memory module
240-pin for DDR2 and DDR3
280-pin for DDR4

Memory controller

Nonvolatile memories

- DRAM and SRAM are volatile memories
  - Lose information if powered off.
- Nonvolatile memories retain value even if powered off
  - Generic name is read-only memory (ROM).
  - Misleading because some ROMs can be read and modified.

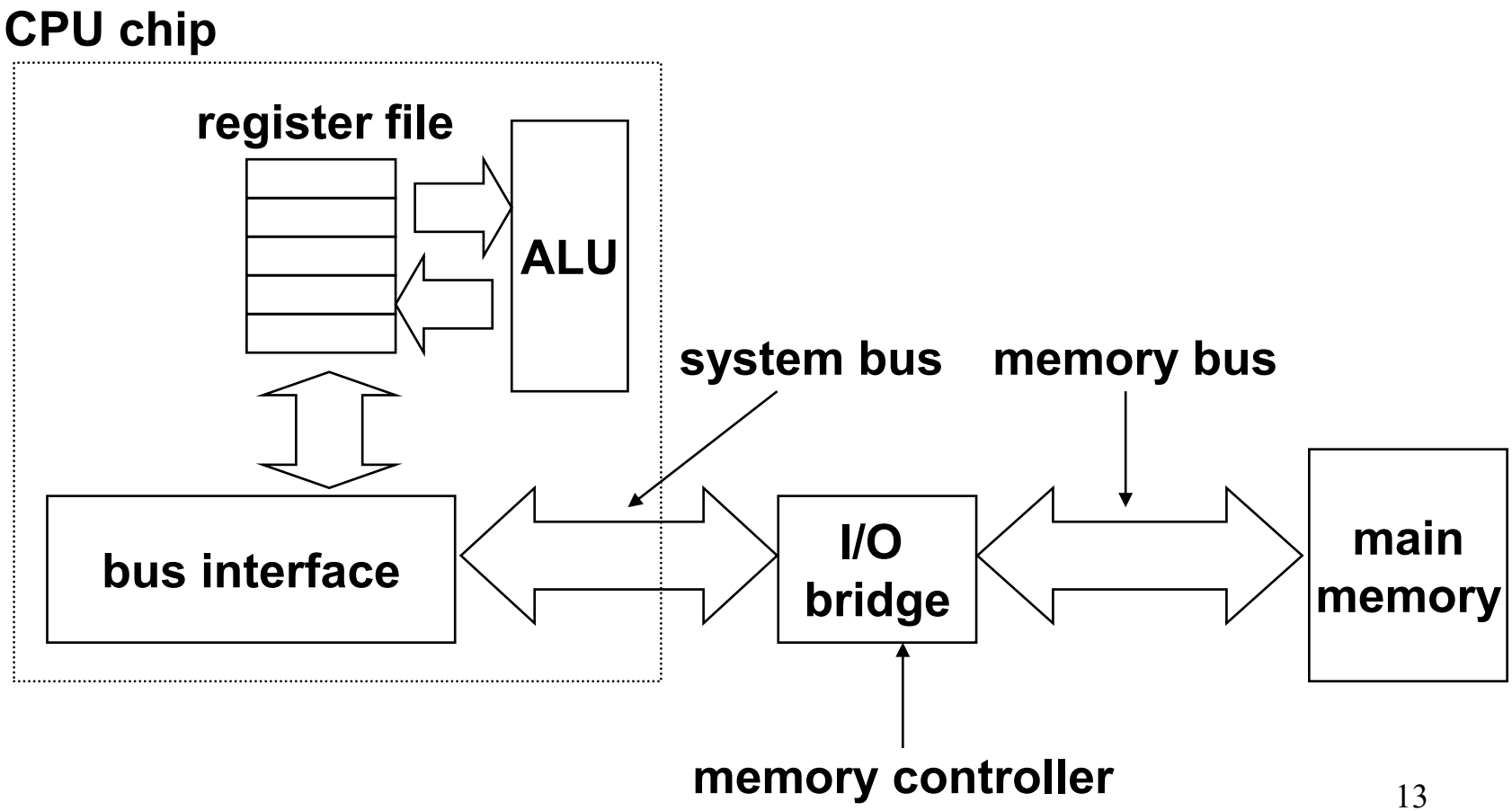
Nonvolatile memories

- Firmware
  - Program stored in a ROM
  - BIOS (basic input/output system)
    - Boot time code
    - a small set of primitive input and output functions
  - graphics cards, disk controllers
    - Translate I/O (input/output) requests from the CPU

Bus Structure Connecting CPU and memory

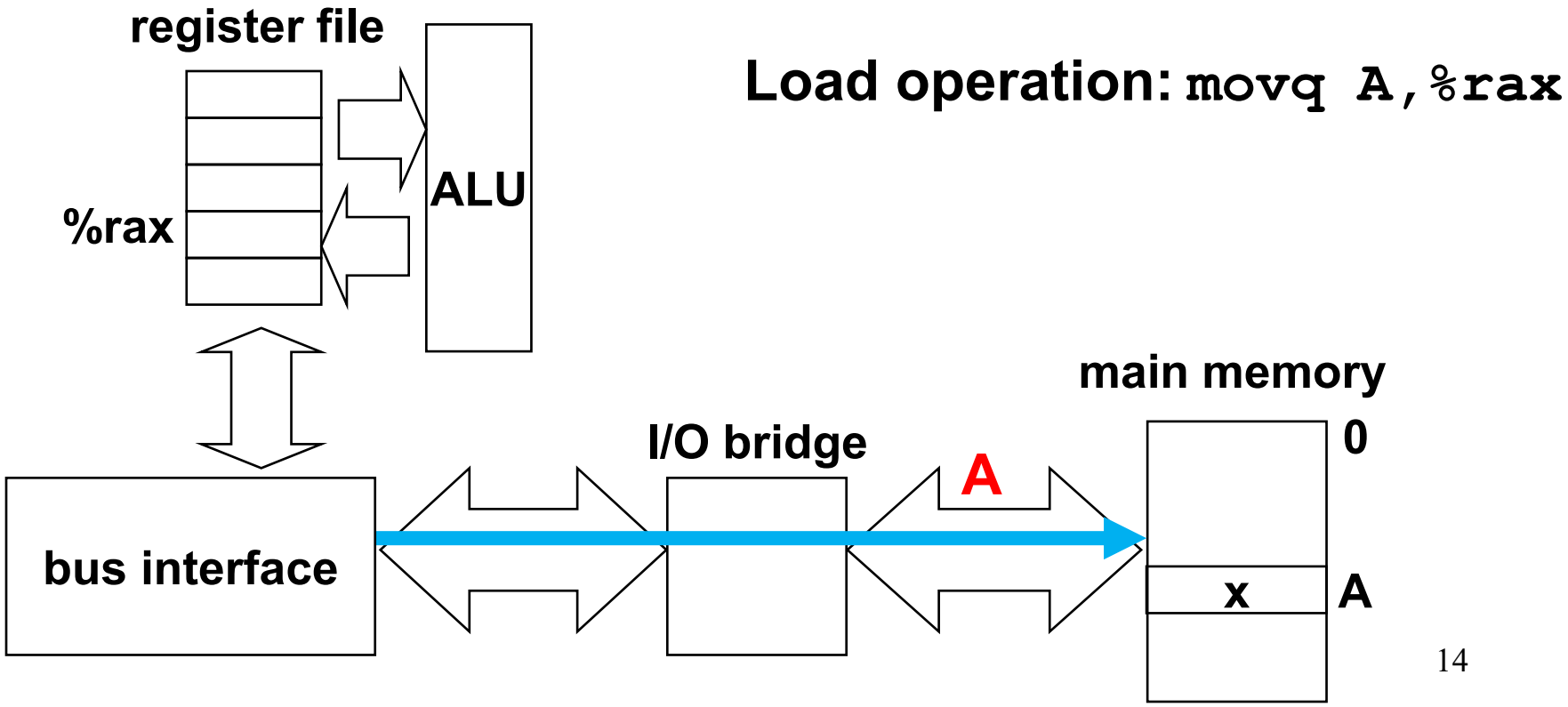
- A **bus** is a collection of parallel wires that carry address, data, and control signals
- Buses are typically shared by multiple devices

Bus Structure Connecting CPU and memory



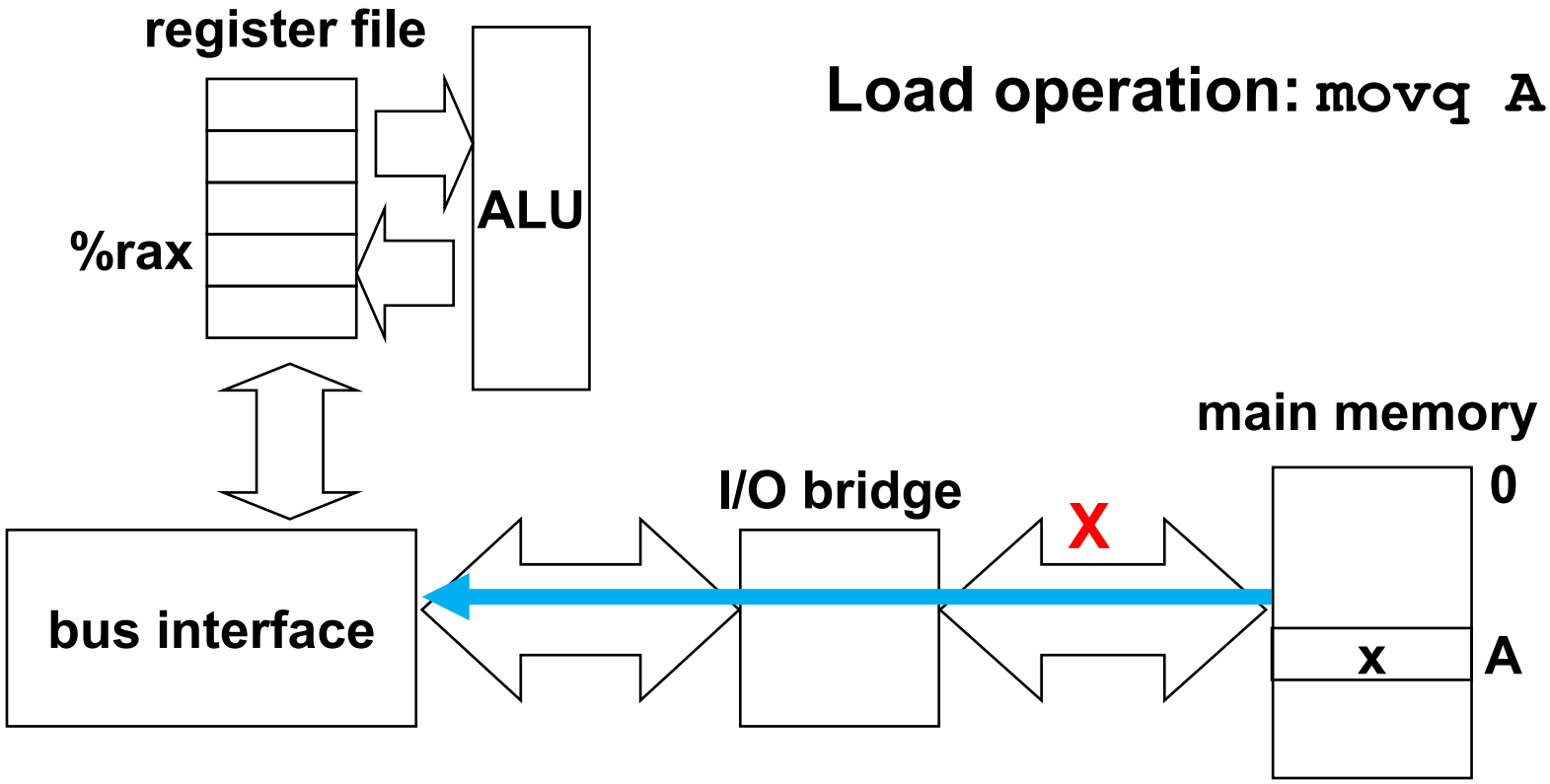
Memory read transaction

1. CPU places address **A** on the memory bus



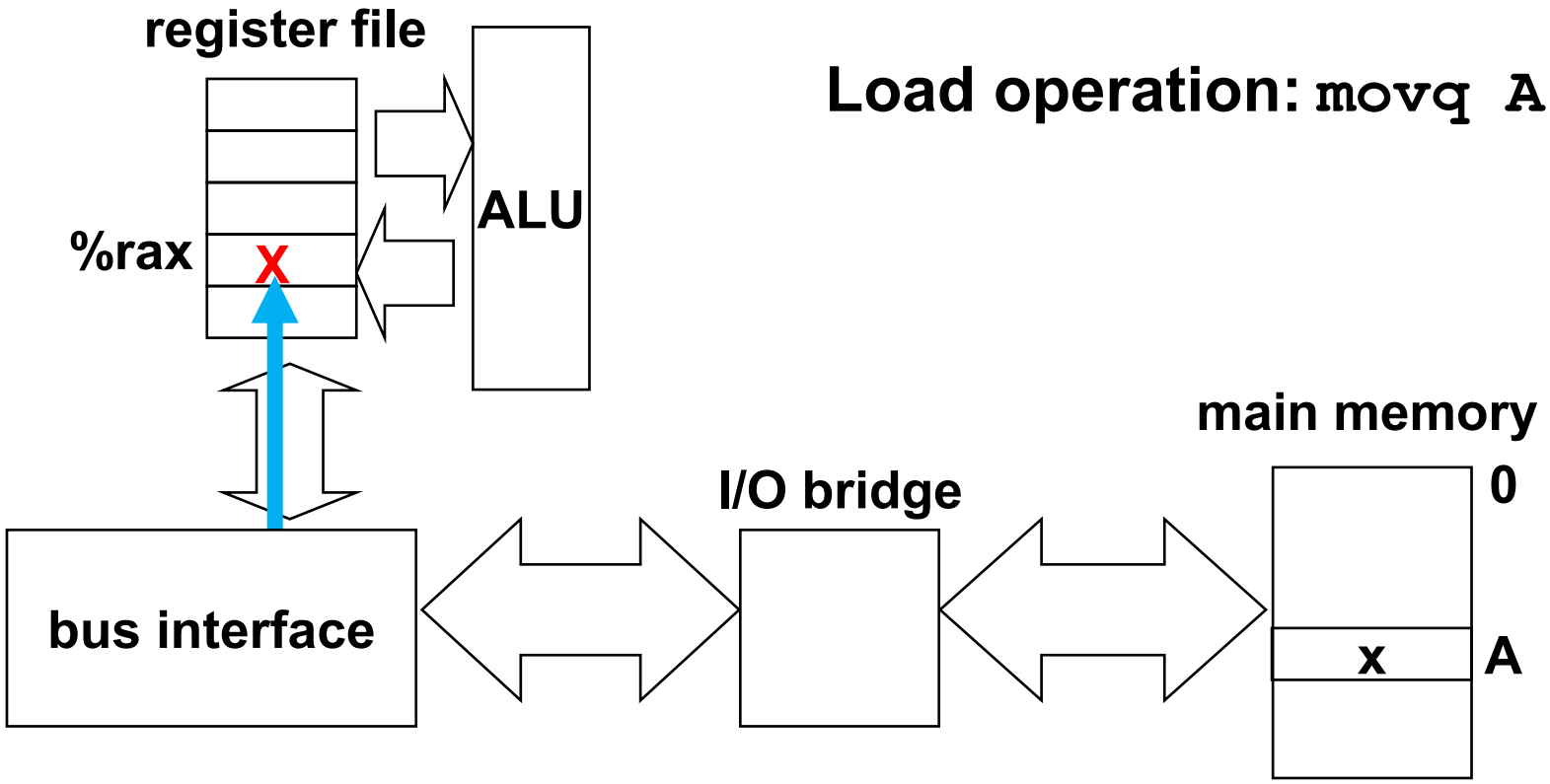
Memory read transaction

2. Main memory reads **A** from the memory bus, retrieves word **x**, and places it on the bus.



Memory read transaction

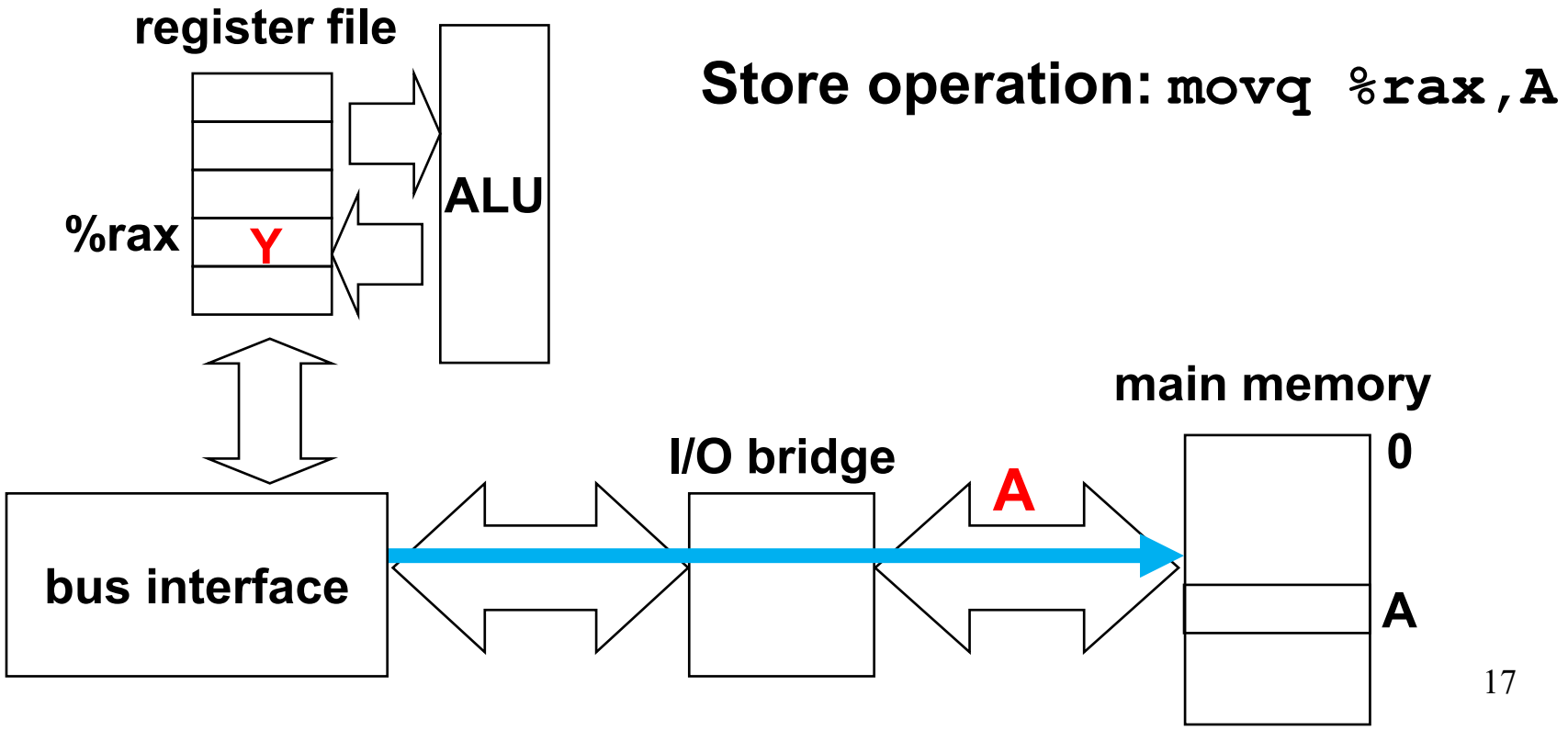
3. CPU read word **x** from the bus and copies it into register **%rax**.



Load operation: `movq A,%rax`

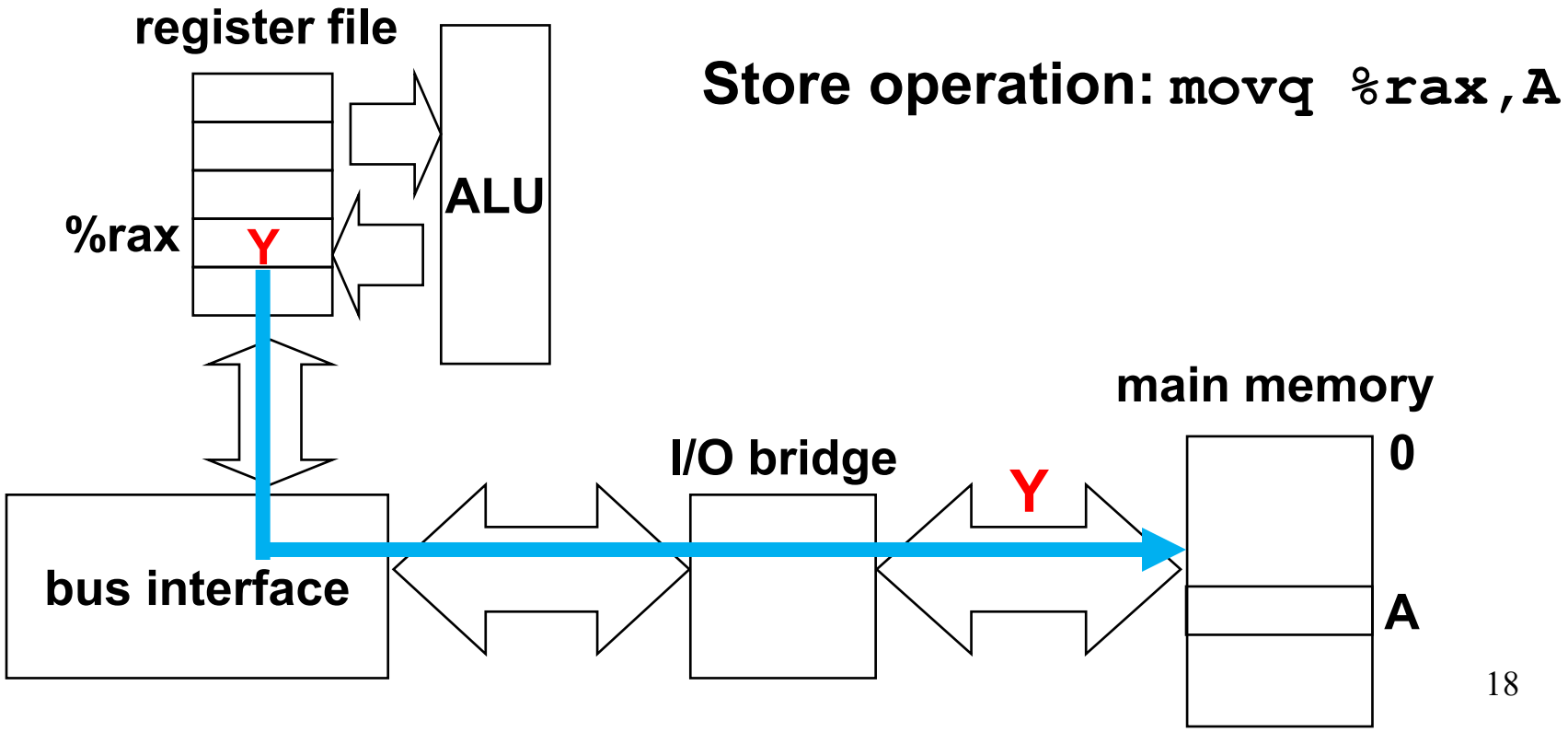
Memory write transaction

1. CPU place address **A** on bus. Main memory reads it and waits for the corresponding data to arrive.



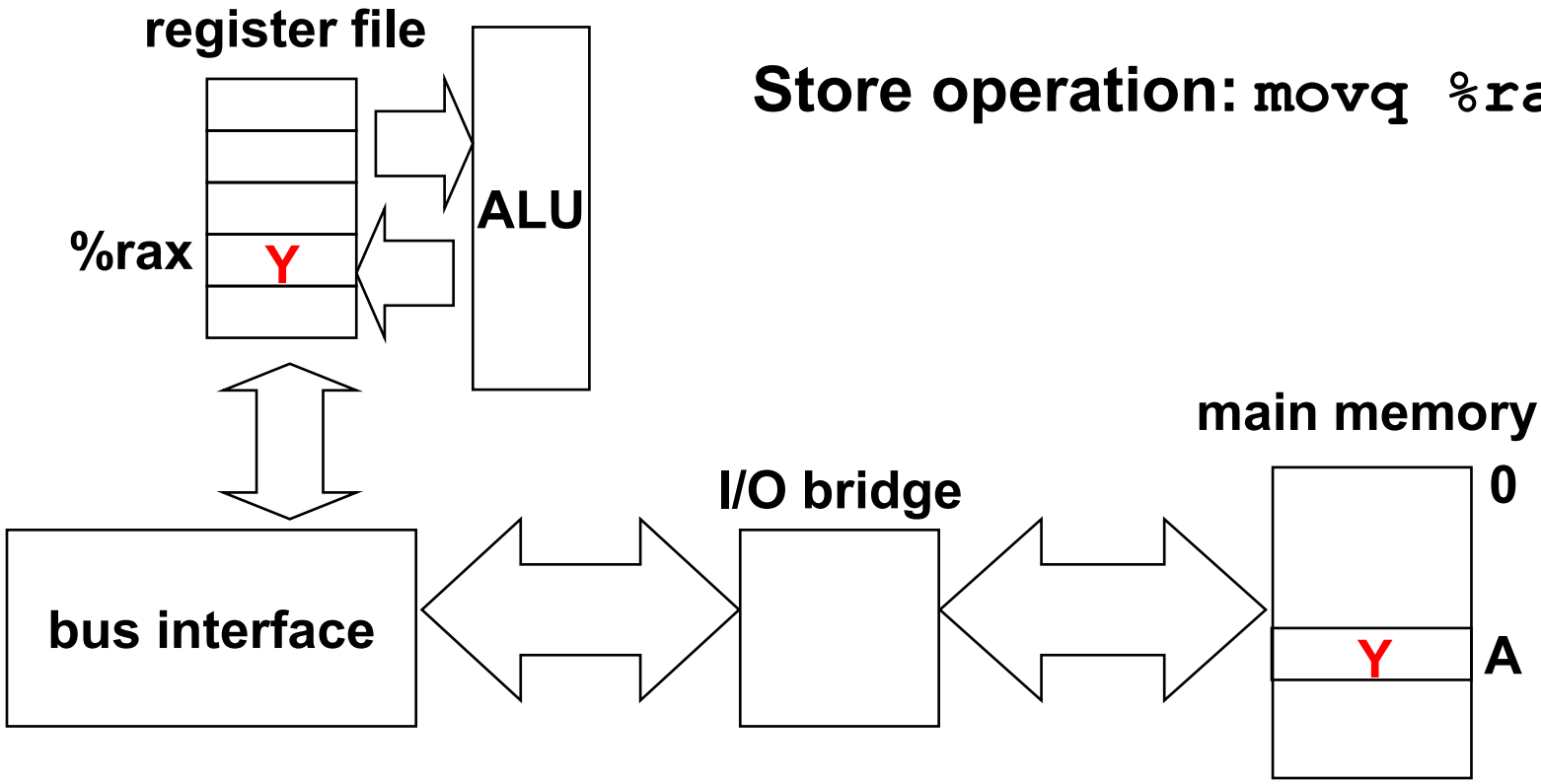
Memory write transaction

2. CPU places data word **Y** on the bus.



Memory write transaction

3. Main memory read data word **Y** from the bus and stores it at address **A**



What's inside a disk drive?

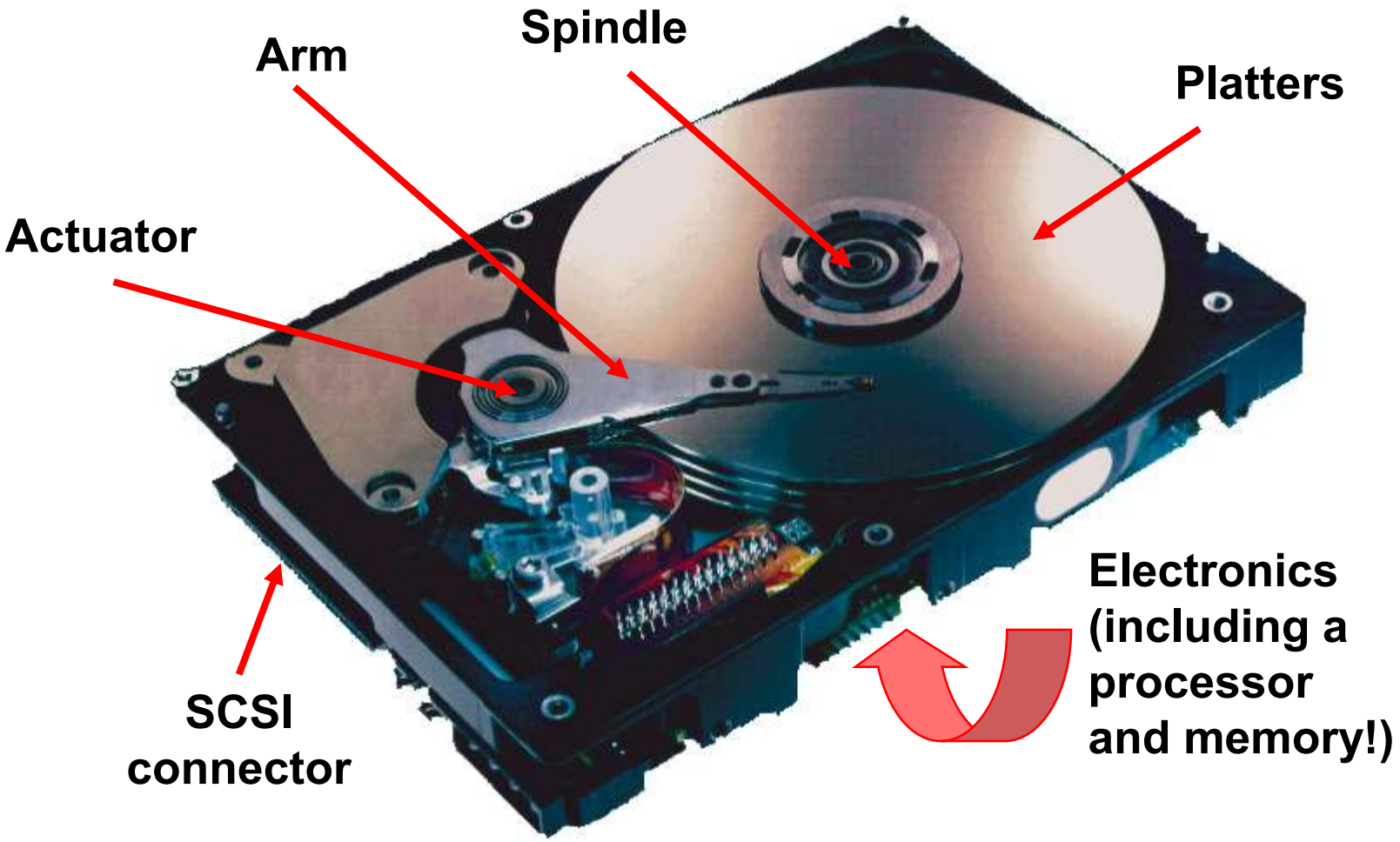
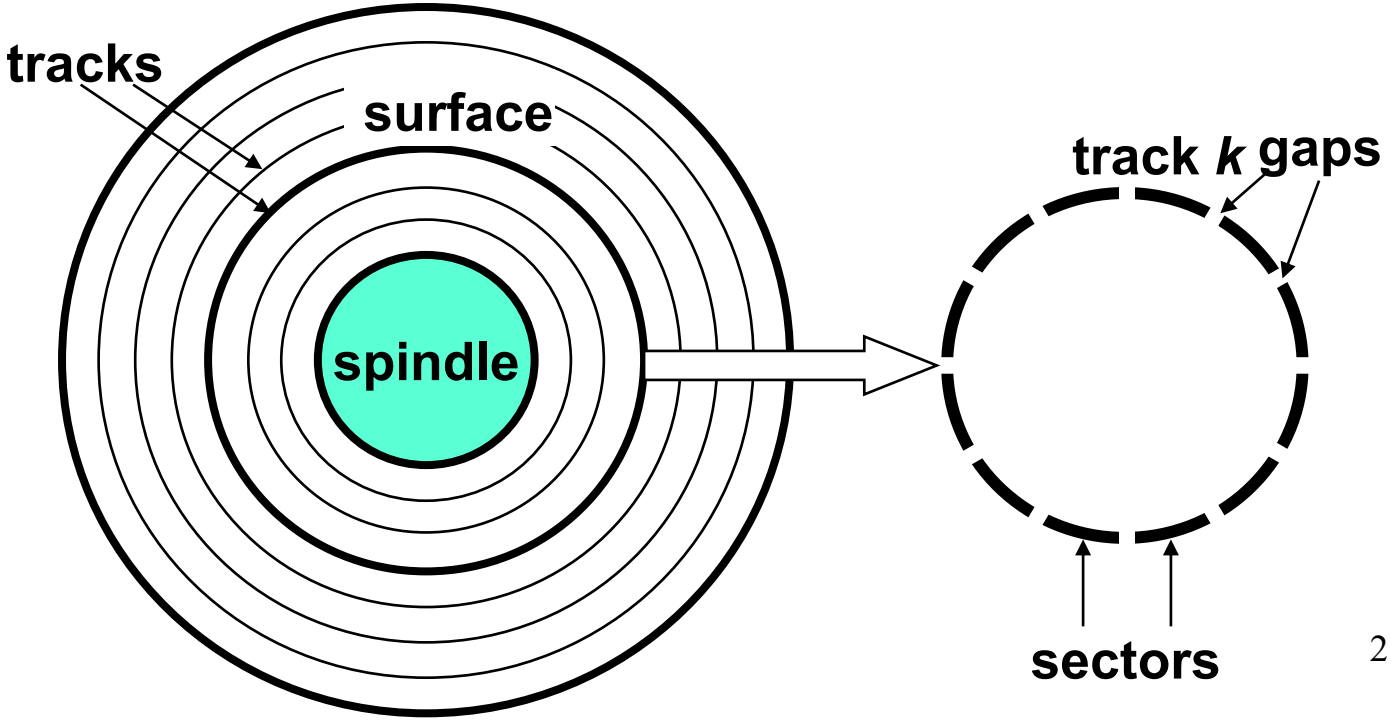


Image courtesy of Seagate Technology

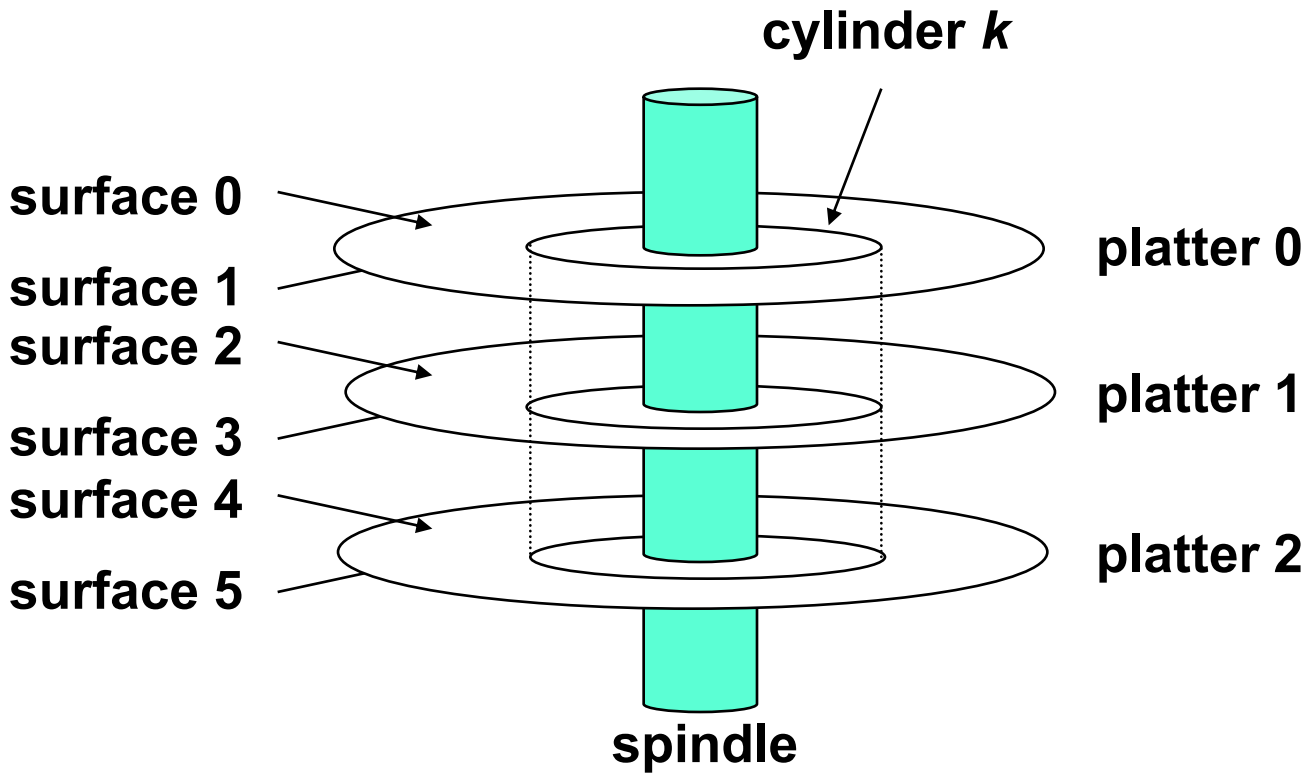
Disk geometry

- Disks consist of **platters**, each with two **surfaces**.
- Each surface consists of concentric rings called **tracks**.
- Each track consists of **sectors** separated by gaps.



Disk geometry (muliple-platter view)

- Aligned tracks form a cylinder.



Disk capacity

- Capacity
  - maximum number of bits that can be stored
  - Vendors express capacity in units of gigabytes (GB), where 1 GB = 10^9.
- Capacity is determined by these technology factors:
  - Recording density (bits/in): number of bits that can be squeezed into a 1 inch segment of a track.
  - Track density (tracks/in): number of tracks that can be squeezed into a 1 inch radial segment.
  - Areal density (bits/in^2): product of recording and track density.

Disk capacity

- Old fashioned disks
  - Each track has the same number of sectors
- Modern disks partition tracks into disjoint subsets called recording zones
  - Each track in a zone has the same number of sectors, determined by the circumference of innermost track
  - Each zone has a different number of sectors/track

Computing disk capacity

Capacity = (# bytes/sector)
x (avg. # sectors/track)
x (# tracks/surface)
x (# surfaces/platter)
x (# platters/disk)

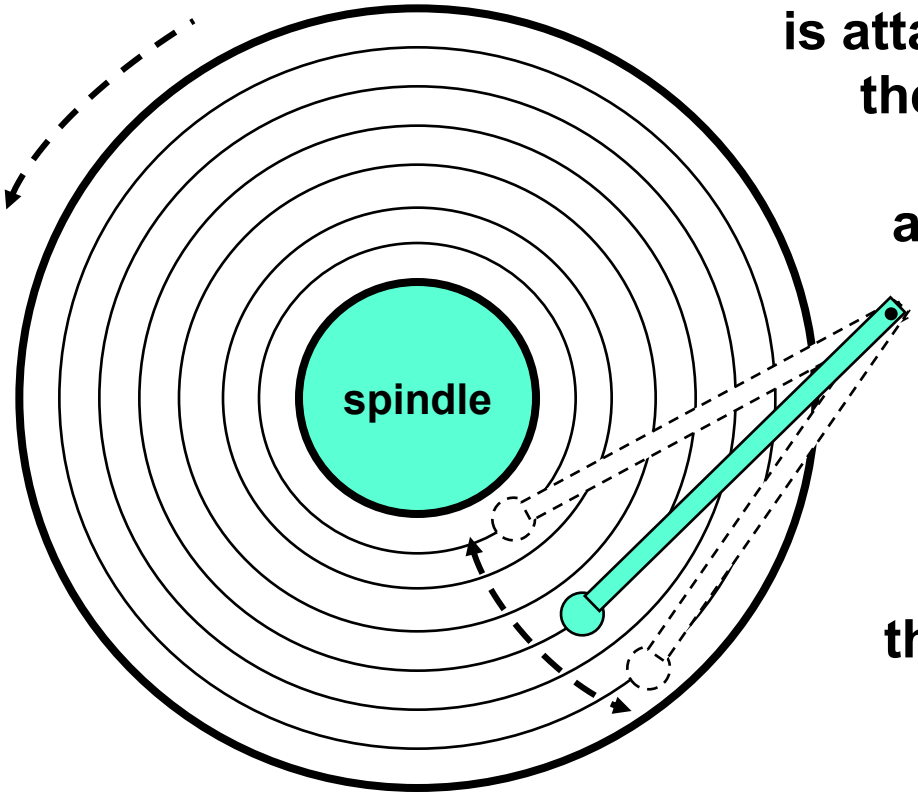
Example:

- 512 bytes/sector
- 300 sectors/track (on average)
- 20,000 tracks/surface
- 2 surfaces/platter
- 5 platters/disk

Capacity = 512 x 300 x 20000 x 2 x 5
= 30,720,000,000
= 30.72 GB

Disk operation (single-platter view)

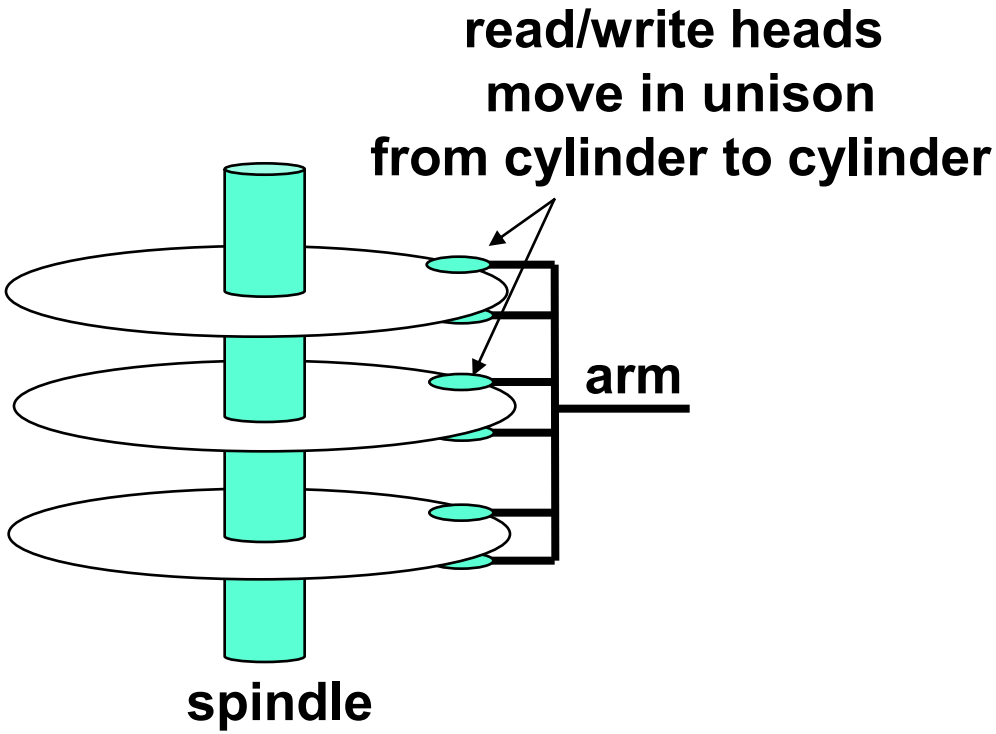
The disk surface spins at a fixed rotational rate



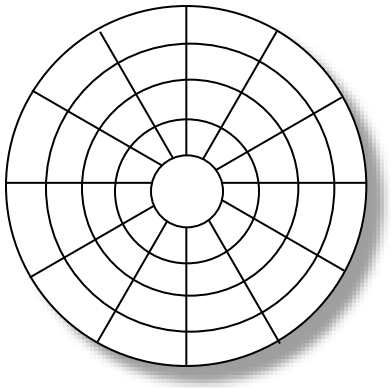
The read/write *head* is attached to the end of the *arm* and flies over the disk surface on a thin cushion of air.

By moving radially, the arm can position the read/write head over any track.

Disk operation (multi-platter view)



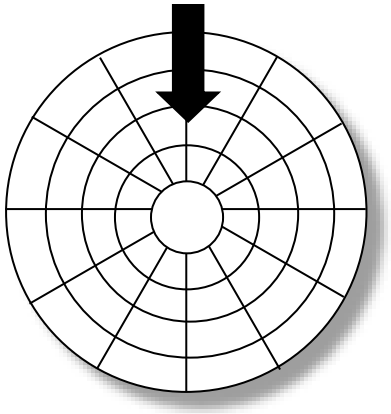
Disk Structure - top view of single platter



Surface organized into tracks

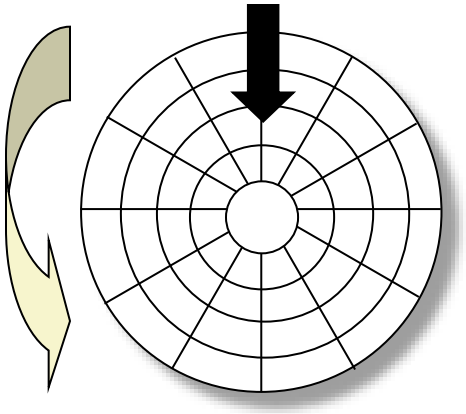
Tracks divided into sectors

Disk Access



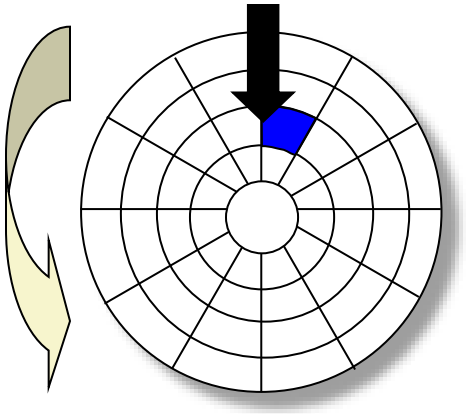
Head in position above a track

Disk Access



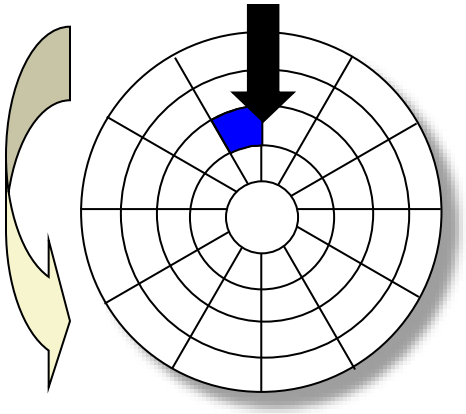
Rotation is counter-clockwise

Disk Access – Read



About to read blue sector

Disk Access – Read

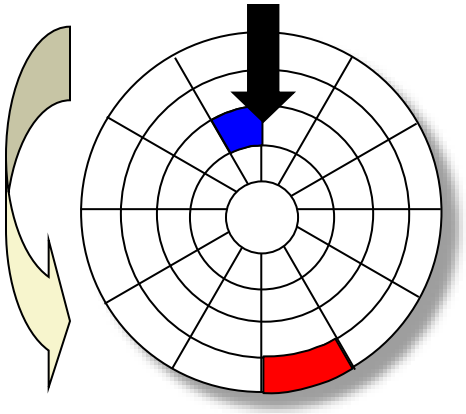


After BLUE read

After reading blue sector

Read one sector each time!

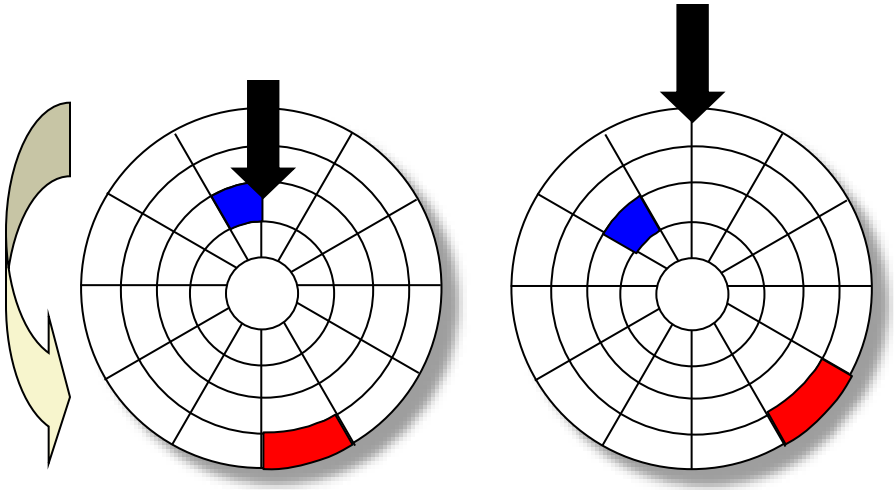
Disk Access – Read



After BLUE read

Red request scheduled next

Disk Access – Seek

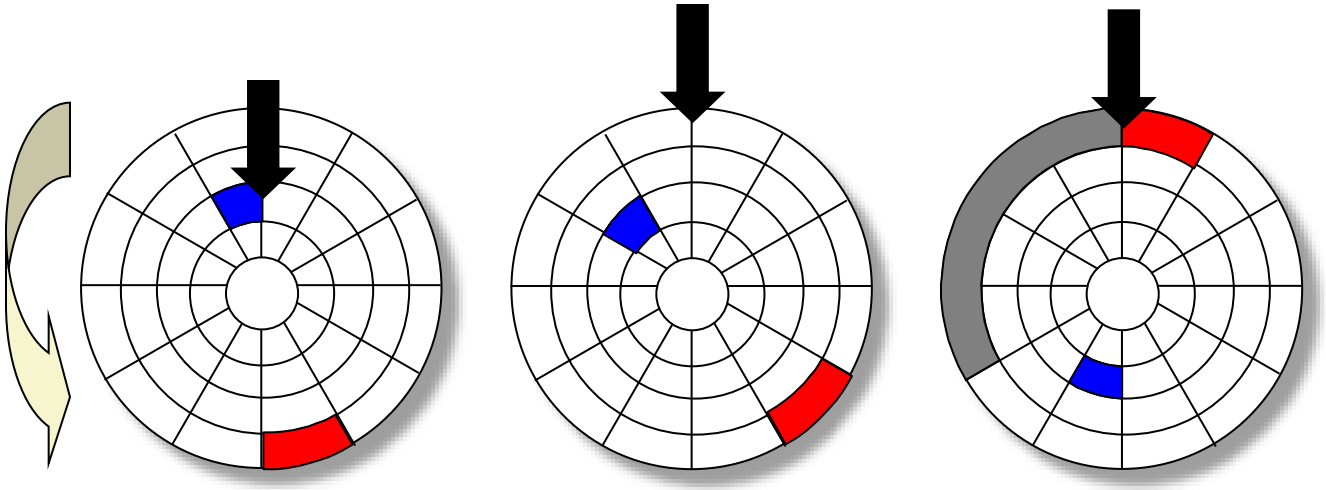


After BLUE read

Seek for RED

Seek to red's track

Disk Access – Rotational Latency



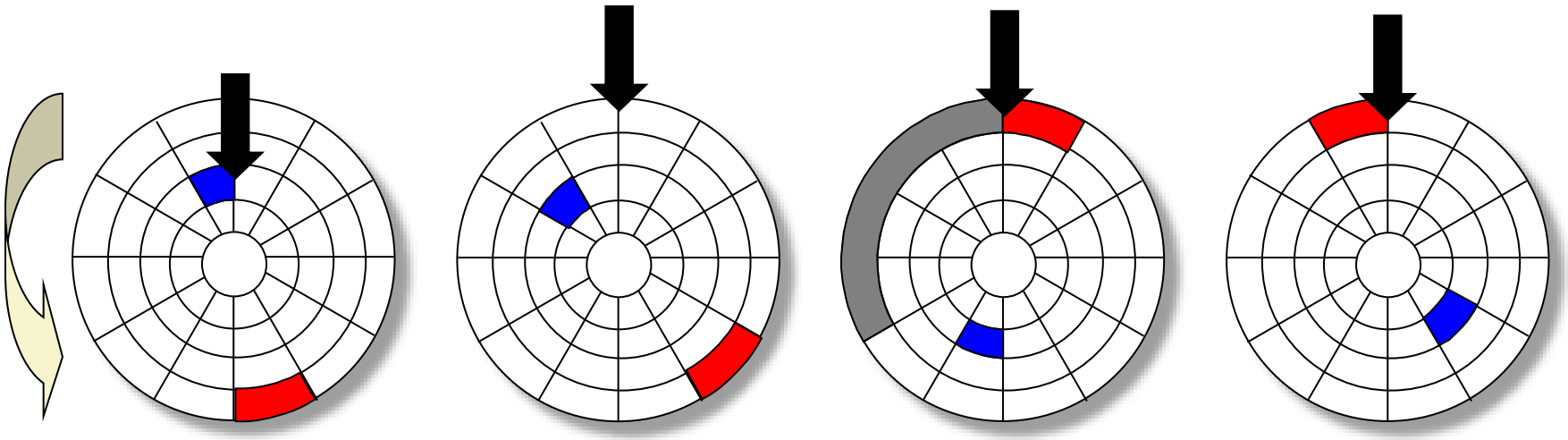
After BLUE read

Seek for RED

Rotational latency

Wait for red sector to rotate around

Disk Access – Read



After BLUE read

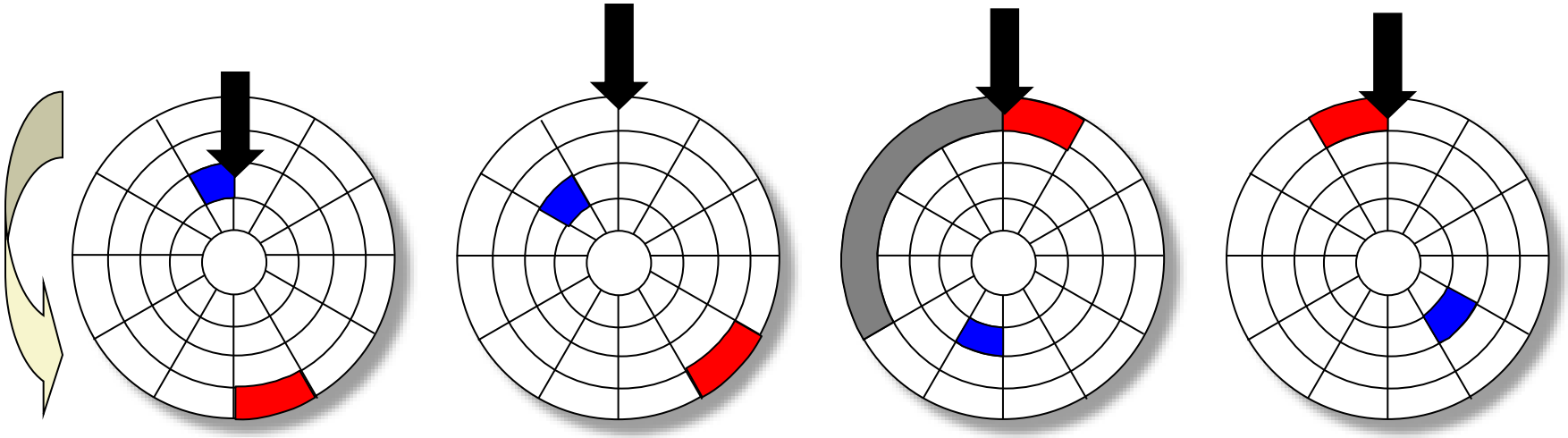
Seek for RED

Rotational latency

After RED read

Complete read of red

Disk Access – Service Time Components



Try to read RED

Seek for RED

Rotational latency

After RED read

Seek

Rotational latency

Data transfer

Disk access time

- Average time to access some target sector approximated by
  - $T_{access} = T_{avg\ seek} + T_{avg\ rotation} + T_{avg\ transfer}$
- Seek time
  - Time to position heads over cylinder containing target sector.
  - Typical $T_{avg\ seek}$ = 9 ms

Disk access time

- Rotational latency
  - Time waiting for first bit of target sector to pass under r/w head.
  - $T_{avg\ rotation}$ = 1/2 x 1/RPMs x 60 sec/1 min
- Transfer time
  - Time to read the bits in the target sector.
  - $T_{avg\ transfer}$ = 1/(avg # sectors/track) x 1/RPM x 60 secs/1 min.

Disk access time example

Example:

Rotational rate 7,200 RPM

$T_{avg\ seek}$ 9 ms

Avg # sectors/track 400

$T_{avg\ rotation}$ = 1/2 * (60secs/7200RPM) * 1000ms/sec

= 4ms

$T_{avg\ transfer}$ = 1/400secs/track x 60sec/7200RPM * 1000ms/sec

= 0.02 ms

T_{access} = 9 ms + 4 ms + 0.02 ms

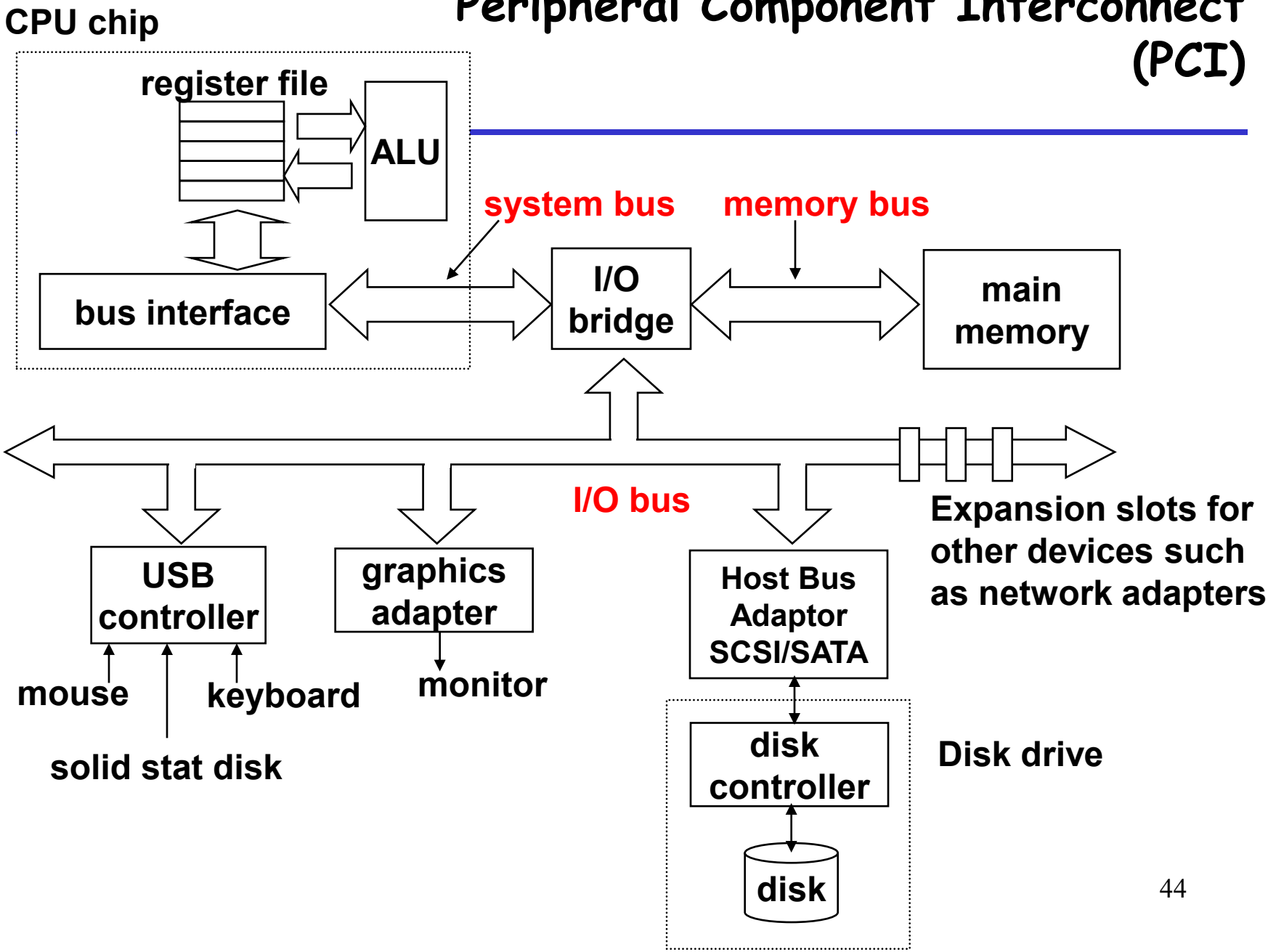
Disk access time example

- Important points
  - Access time dominated by seek time and rotational latency, about 10ms for 512 bytes
  - First bit in a sector is the most expensive, the rest are free
  - SRAM access time is about 4 ns/64-bit word
  - DRAM is about 60 ns/64-bit word
  - Disk is about 40,000 times slower than SRAM
  - Disk is about 2,500 times slower then DRAM

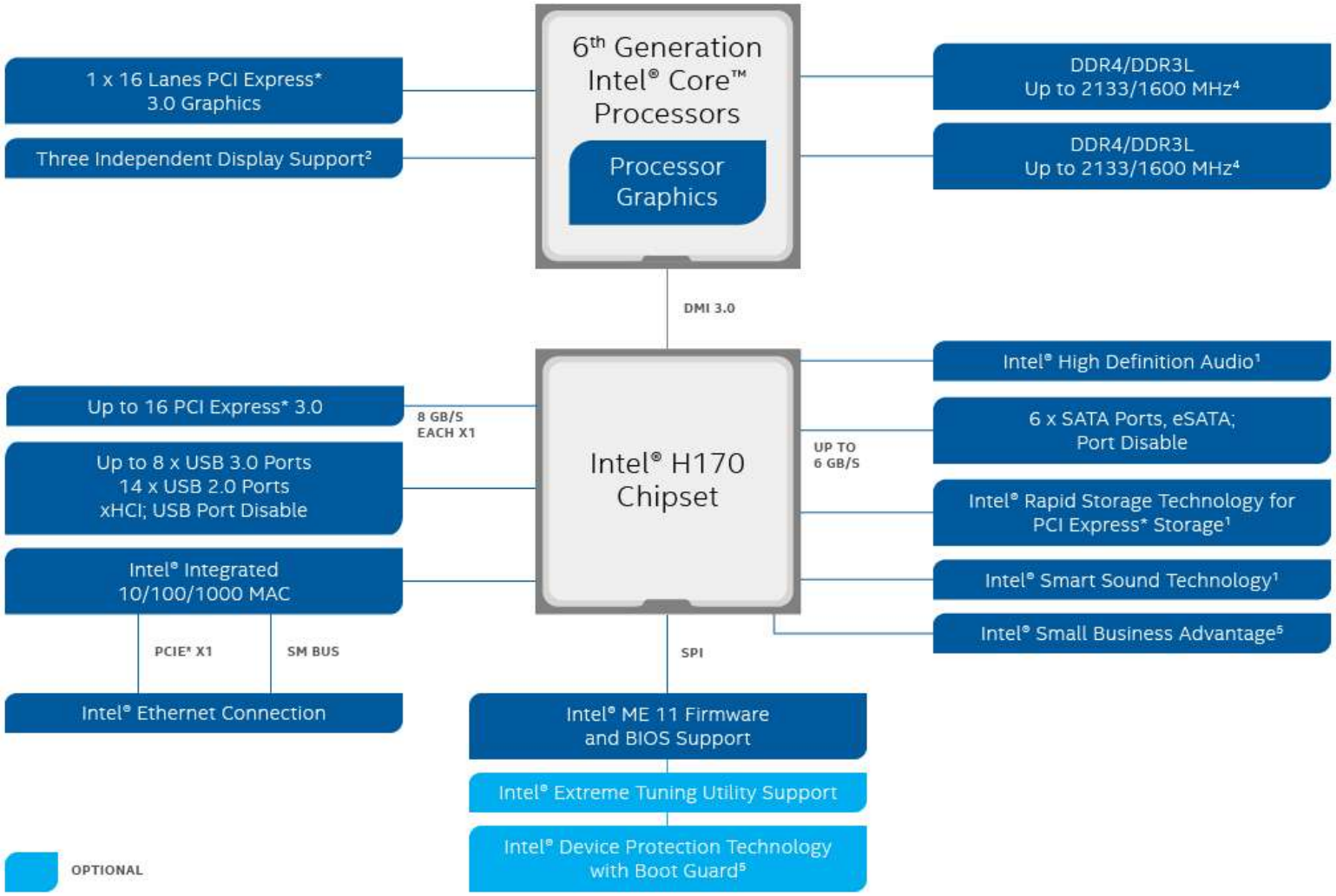
Logical disk blocks

- Modern disks present a simpler abstract view of the complex sector geometry:
  - The set of available sectors is modeled as a sequence of B sector-size logical blocks (0, 1, ...,B-1)
- Mapping between logical blocks and actual (physical) sectors
  - Maintained by hardware/firmware device called disk controller
  - Converts requests for logical blocks into (surface, track, sector) triples.

Peripheral Component Interconnect (PCI)



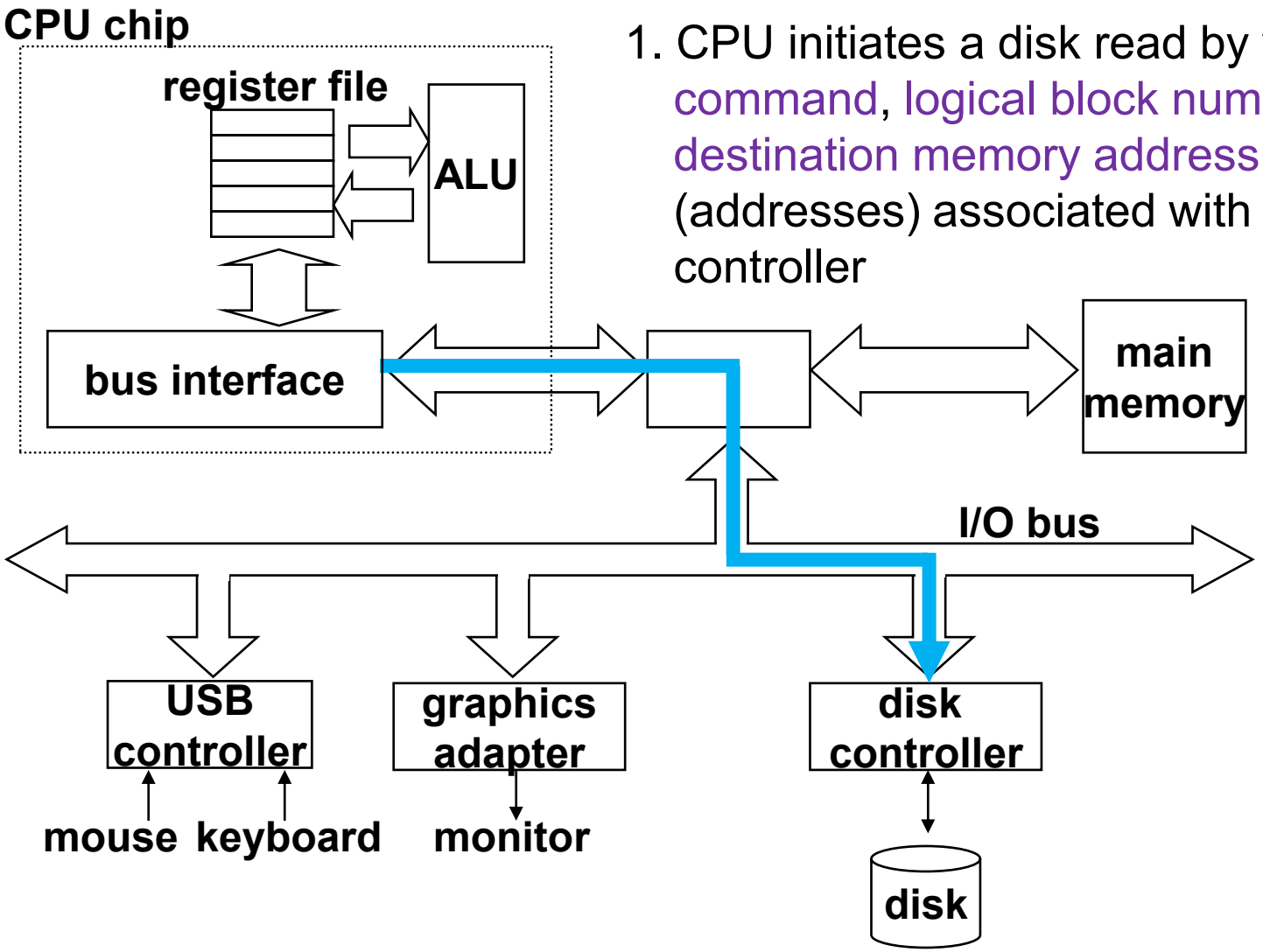
INTEL® H170 CHIPSET BLOCK DIAGRAM



Memory-mapped I/O

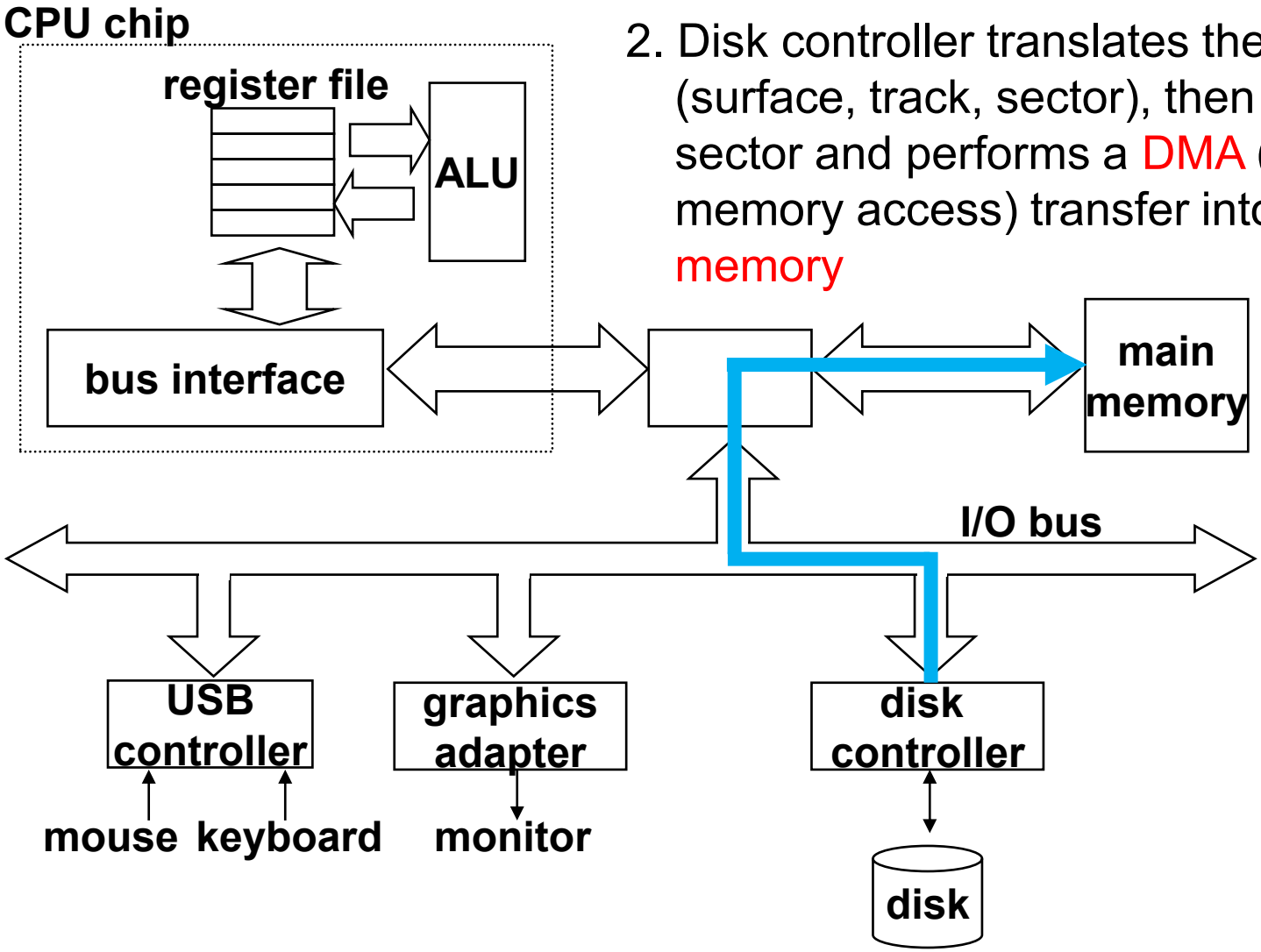
- There are processors in controllers and adapters
- I/O Port
  - A reserved address in the address space
  - Some registers in the above processors are mapped to these addresses
- Each device is associated with
  - One or more ports when it is attached to the bus
- The CPU is able to communicate with an I/O device via its I/O ports

Reading a disk block



1. CPU initiates a disk read by writing a command, logical block number, and destination memory address to ports (addresses) associated with the disk controller

Reading a disk sector

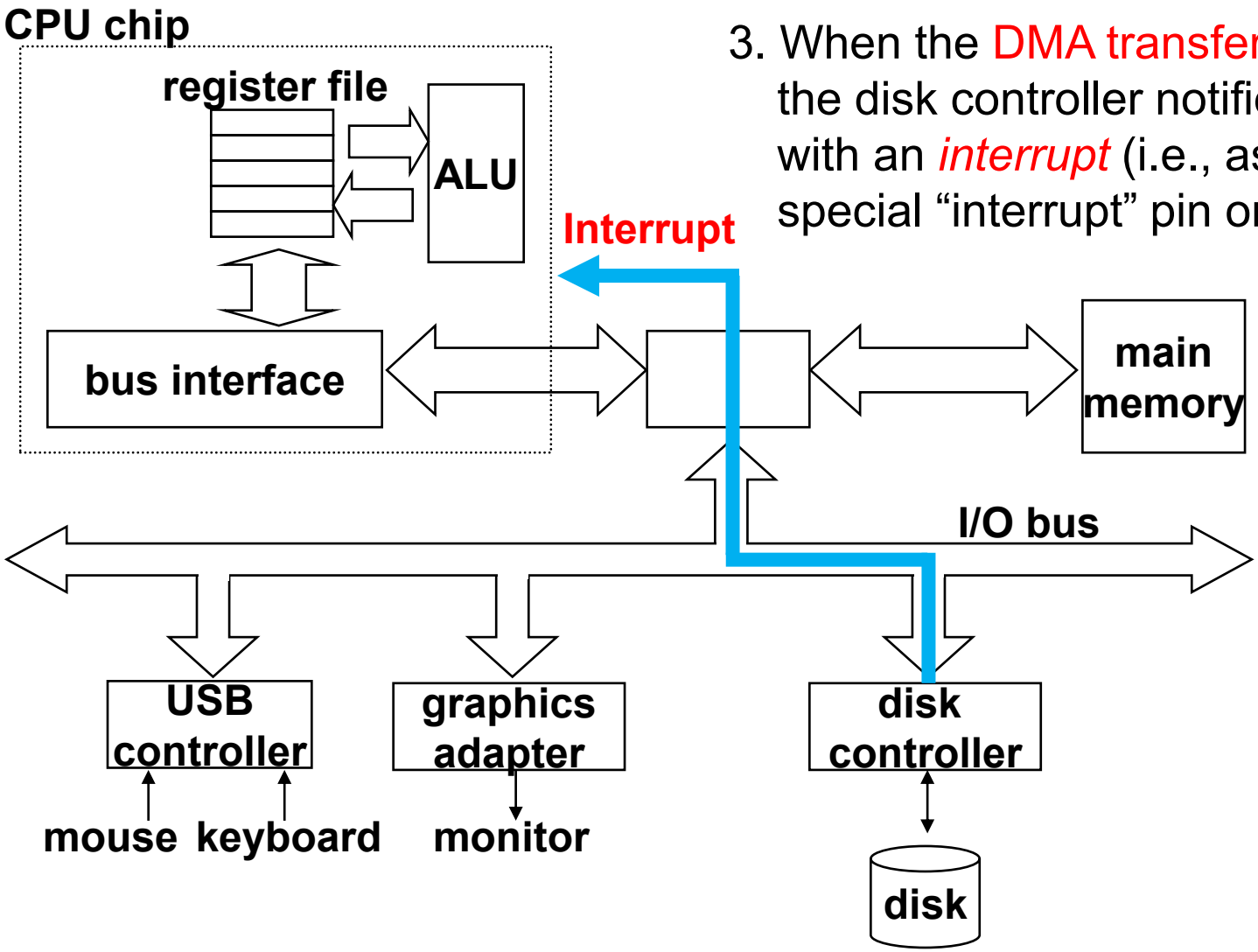


2. Disk controller translates the block to (surface, track, sector), then reads the sector and performs a DMA (direct memory access) transfer into main memory

DMA

- Direct memory access
- A device performs a read or write bus transaction on its own, without any involvement of the CPU
- The transfer of data is known as a DMA transfer

Reading a disk block



3. When the DMA transfer completes, the disk controller notifies the CPU with an *interrupt* (i.e., asserts a special "interrupt" pin on the CPU)

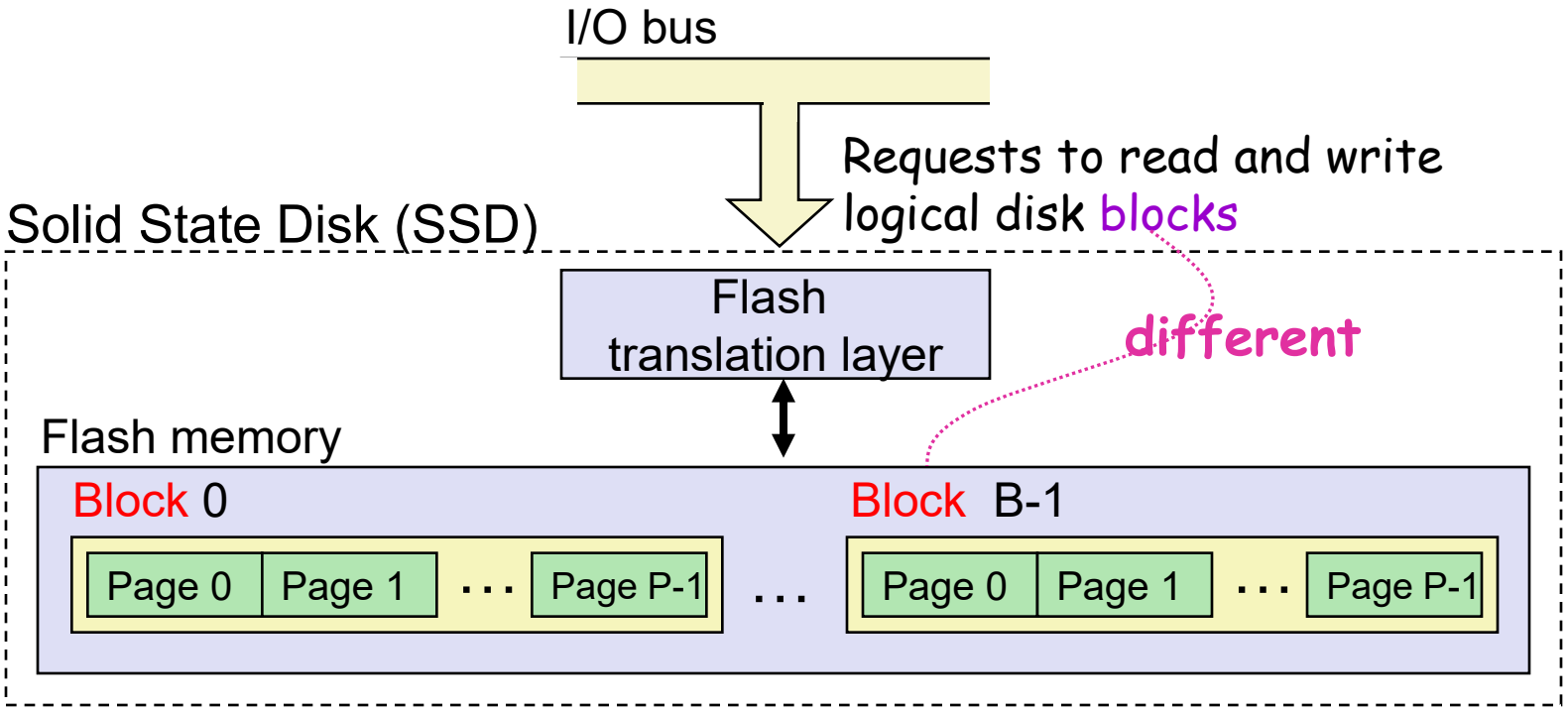
Interrupt

- I/O devices trigger interrupts by signaling a pin on the processor chip
  - network adapters, disk controllers, timer chips
- I/O devices place a number onto the system bus
  - identifies the device that caused the interrupt
- Cause the CPU stop what it is currently working on, and jump to an operating system routine

Solid State Disk (SSD)

- Solid state disk (SSD) is a storage technology, based on flash memory
  - SSD package plugs into a standard disk slot on the I/O bus (typically USB or SATA or PCIe)
  - Flash translation layer = disk controller

Solid State Disk (SSD)



- Pages: 512B to 4KB, Blocks: 32 to 128 pages
- Data read/written in units of pages.
- Page can be written only after its block has been erased
- A block wears out after 100,000 repeated writes.

SSD Performance Characteristics

- Benchmark of Samsung 970 EVO Plus

https://ssd.userbenchmark.com/SpeedTest/711305/Samsung-SSD-970-EVO-Plus-250GB

Sequential read throughput	**2,126 MB/s**	**Sequential write tput**	**1,880 MB/s**
Random read throughput	**140 MB/s**	**Random write tput**	**59 MB/s**

- Sequential access faster than random access
  - Common theme in the memory hierarchy
- Random writes are somewhat slower
  - Erasing a block takes a long time (~1 ms).
  - Modifying a block page requires all other pages to be copied to new block.
  - Flash translation layer allows accumulating series of small writes before doing block write.

SSD Tradeoffs vs Rotating Disks

- Advantages
  - No moving parts → faster, less power, more rugged
- Disadvantages
  - Have the potential to wear out
    - Mitigated by "wear leveling logic" in flash translation layer
    - E.g. Samsung 970 EVO Plus guarantees 600 writes/byte of writes before they wear out
    - Controller migrates data to minimize wear level
  - In 2019, about 4 times more expensive per byte
    - And, relative cost will keep dropping
- Applications
  - Smartphones, laptops
  - Increasingly common in desktops and servers

Storage Trends

SRAM

Metric	1985	1990	1995	2000	2005	2010	2015	2015:1985
$/MB	2,900	320	256	100	75	60	25	116
access (ns)	150	35	15	3	2	1.5	1.3	115

DRAM

Metric	1985	1990	1995	2000	2005	2010	2015	2015:1985
$/MB	880	100	30	1	0.1	0.06	0.02	44,000
access (ns)	200	100	70	60	50	40	20	10
typical size (MB)	0.256	4	16	64	2,000	8,000	16,000	62,500

Disk

Metric	1985	1990	1995	2000	2005	2010	2015	2015:1985
$/GB	100,000	8,000	300	10	5	0.3	0.03	3,333,333
access (ms)	75	28	10	8	4	3	3	25
typical size (GB)	0.01	0.16	1	20	160	1,500	3000	1,500,000

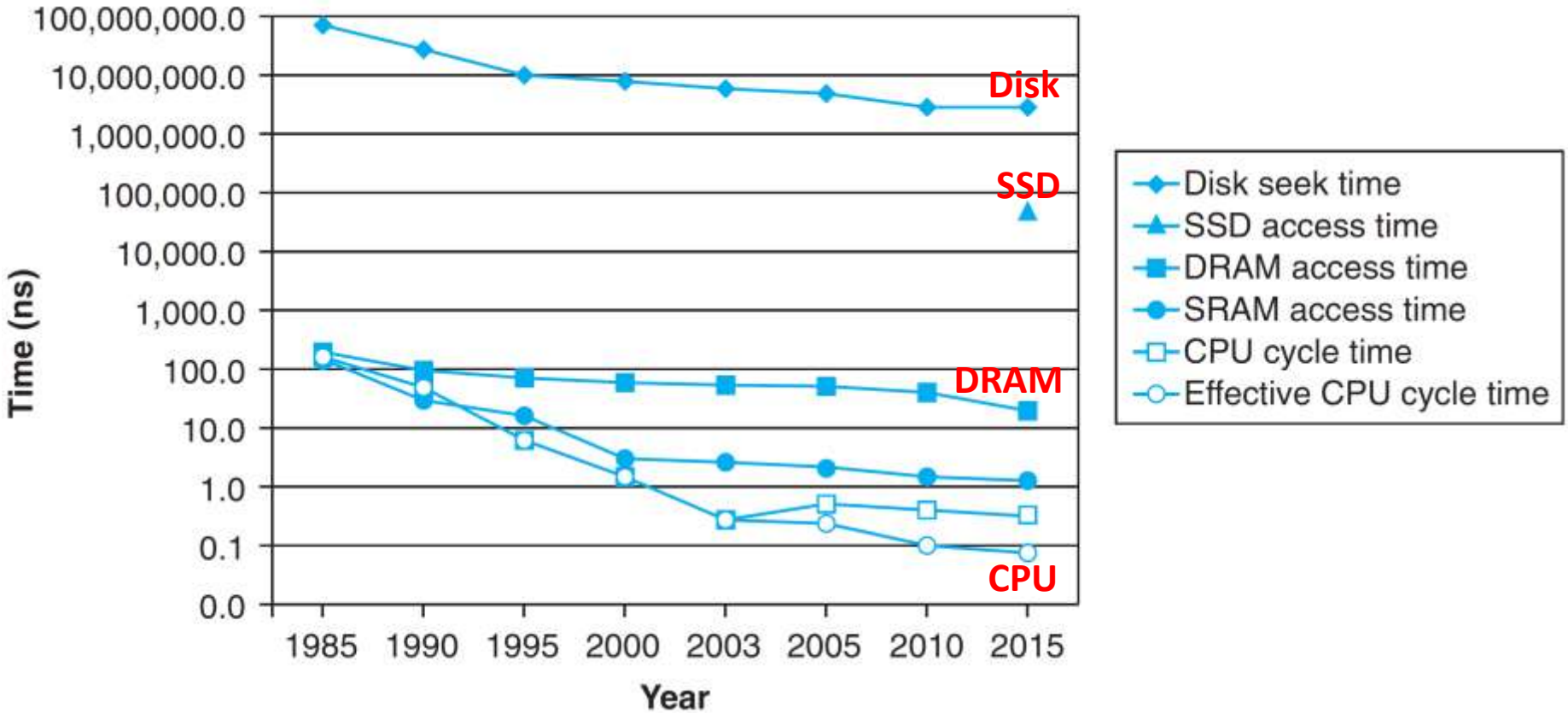
CPU Clock Rates

Inflection point in computer history when designers hit the "Power Wall"

	1985	1990	1995	2000	2003	2005	2010	2015	2015:1985
CPU	80286	80386	Pent.	P-III	Pent.4	Core 2	Core i7(n)	Core i7(h)	---
Clock rate (MHz)	6	20	150	600	3,300	2,000	2,500	3,000	500
Cycle time (ns)	166	50	6	1.6	0.3	0.5	0.4	0.33	500
Cores	1	1	1	1	1	2	4	4	4
Effective cycle time (ns)	166	50	6	1.6	0.3	0.25	0.10	0.08	2,075

The CPU-Memory Gap

The gap **widens** between DRAM, disk, and CPU speeds.



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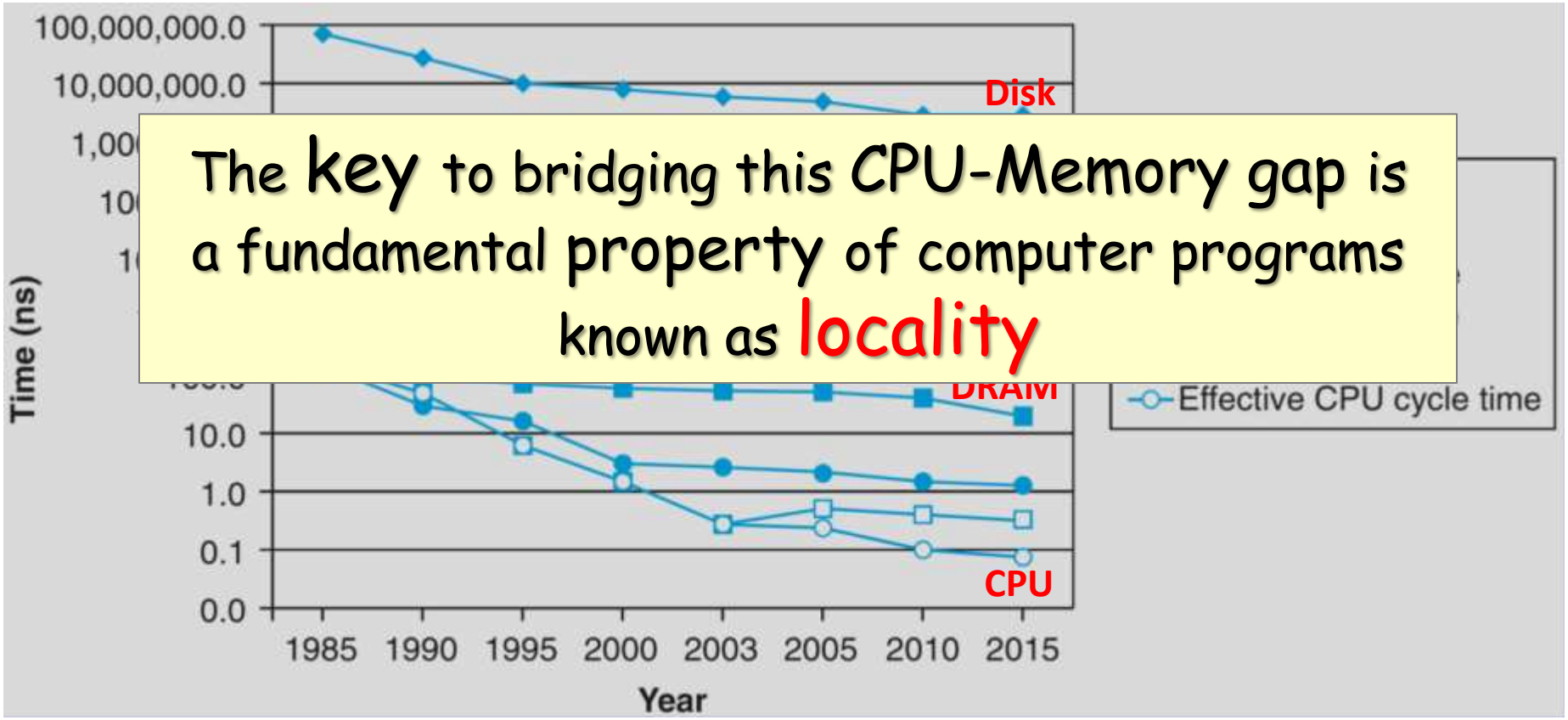

The key to bridging this CPU-Memory gap is a fundamental property of computer programs known as locality